

SRI CHARAN SRIPADI

AI Engineer | Agentic AI, LLM Evaluation and MLOps

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PROFESSIONAL SUMMARY

MSc student in Advanced AI and Data Science at the University of Liverpool. I build multi-agent systems on Google ADK and fine-tune small models to evaluate them, then deploy the results with Docker and CI. My recent open-source projects address open feature requests in Google's Agent Development Kit and AWS Strands Agents, and I report each result on held-out data along with its limits. I work mainly in Python, PyTorch and Hugging Face Transformers, and I have contributed to the Hugging Face Transformers repository (PR #46777). Eligible to work in the UK without sponsorship.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C, C++

Agentic AI and LLMs: Google ADK, multi-agent systems, tool calling, Agent Skills (SKILL.md), LLM-as-judge, agent simulation and evaluation, RAG, LiteLLM, Groq, prompt engineering

ML and fine-tuning: PyTorch, TensorFlow, Hugging Face Transformers, ModernBERT, scikit-learn, calibration (Platt scaling, ECE), A/B testing (McNemar), clustering, OpenCV, Dlib, NLP

MLOps and serving: MLflow, Hugging Face Hub (model registry), FastAPI, Flask, REST APIs, Prometheus metrics, data versioning, Docker, GitHub Actions, GHCR, Google Colab GPUs

Cloud and data: AWS (EC2, S3), Google Cloud Platform, Microsoft Azure, Vercel, MySQL, SQLite, FAISS, Chroma

Engineering: Pytest (unit, integration and offline model tests), Git, Linux, OOP, system design, Agile/Scrum

EDUCATION

MSc Advanced Artificial Intelligence and Data Science, University of Liverpool, UK | 2025 - Present

BTech in Artificial Intelligence and Data Science, KL University, India | 2021 - 2025 | GPA 9.4 / 10.0, First Class with Distinction

EXPERIENCE

Independent AI Engineer: Open-Source Agent Tooling and ML Systems | 2023 - Present

- Built and open-sourced SkillMiner and TinyJudge, two agent infrastructure projects that address open feature requests in Google ADK and AWS Strands Agents. I handled everything from data generation and fine-tuning to deployment with Docker.
- Built AI systems for computer vision, NLP and multimodal agent pipelines, served through Flask and FastAPI REST APIs under 300 ms, with CI/CD on GitHub Actions.
- Wrote Pytest suites (unit, integration, and offline tests that use small stand-in models) so the pipelines run in CI without API keys or model downloads.

Open Source Contributor, Hugging Face Transformers | 2025 - Present

- Submitted PR #46777 to the Hugging Face Transformers repository. It replaces wildcard imports with explicit named imports in the TimeSformer model module and resolves issue #45306. github.com/huggingface/transformers/pull/46777

PROJECTS

TinyJudge: Fine-tuned, Calibrated Judge for AI Agent Behaviour | github.com/Charansripadi/tinyjudge

PyTorch · Hugging Face Transformers · ModernBERT · MLflow · FastAPI · Docker · GitHub Actions · HF Hub

- Fine-tuned ModernBERT-base (150M parameters) to judge AI agents on policy compliance, tool choice, and whether replies match the tool results. It is an open alternative to LLM-as-judge, as requested in AWS Strands Agents issue #4585.
- Generated a versioned training set (SHA-256 manifest) from rule-labelled agent trajectories and LLM-written replies. On 208 held-out conversations from a real LLM agent it scored 88.0% accuracy, against 81.2% for a 27B LLM judge (Qwen 3.8).
- Fitted Platt calibration per task on a separate half of the real data. Calibration error fell from 0.20 to 0.10 (ECE) and faithfulness accuracy rose from 74% to 84%. A confidence cascade reached 89.9% while sending 35% of cases to the LLM.
- Tracked runs in MLflow, trained on a Colab T4 in about 4 minutes, and versioned models on the Hugging Face Hub. The FastAPI service exposes Prometheus metrics and drift signals. CI builds the Docker image, smoke-tests it, and blocks publishing to GHCR if accuracy drops below 85%.

SkillMiner: Multi-Agent System that Mines Agent Traces into Skills | github.com/Charansripadi/skillminer

Google ADK · Multi-agent · LiteLLM · Groq · scikit-learn · Pytest

- Built a multi-agent pipeline on Google ADK 2.10 that finds the workflows an agent repeats across sessions and writes them as Agent Skills (SKILL.md) that ADK loads directly. It addresses Google ADK issue #5398.
- Wrote an ADK plugin that records traces, an LLM customer simulator with personas and hidden labels, and an LLM intent miner. The miner reached purity 1.00 and ARI 0.93, compared with 0.48 for a TF-IDF clustering baseline.
- Added PII redaction based on values seen in tool calls, a paired A/B validator (exact McNemar test) and a drift check for changed tools. The mined refund skill picked up the rule "never refund an ineligible order" from violations in the traces. In a pilot A/B test (8 pairs), compliance went from 75% to 100% and violations from 1 to 0, which is not yet statistically significant.
- Fixed a bug where retrying a rate-limited agent turn ran refund and email tools a second time. Retries now happen per model call. The project has 23 offline tests.

FusionNet: Multimodal Agentic Video Analysis System | github.com/Charansripadi

Python · PyTorch · NLP · Agentic Pipelines · Docker · RAG

- Designed a multimodal pipeline in which four agents (Loader, Vision, Audio, NLP) work under a FusionController, with inter-agent latency under 200 ms.
- Produces JSON summaries, tags, and emotion and energy estimates, with 91% label consistency over 50+ test videos. Containerised with Docker and covered by Pytest integration tests.

Other projects: Driver Drowsiness Detection, a real-time OpenCV and Dlib web app with 95% detection accuracy over 200+ sessions, a 24 FPS Flask API and a 35% latency reduction, deployed with Docker and GitHub Actions. Speech Emotion Recognition, which classifies 7 emotions with 88% accuracy on 10,000 samples, runs inference in under 300 ms and is deployed on AWS EC2 with Docker.

CERTIFICATIONS

- TensorFlow Developer Certification, Google (2024)
- Google Cloud Associate Cloud Engineer (2024)